

SALO KARKKA, Finland

WMO: 027560

Lat: 60.374N

Lon: 23.113E

Elev: 2

StdP: 101.30

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -22.4	(c) -18.4	(d) -24.9	(e) 0.4	(f) -22.3	(g) -20.9	(h) 0.6	(i) -18.2	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	27.9	19.1	25.8	18.1	24.0	17.1	20.6	25.3	19.3	24.0	18.2	22.7	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.8	23.0	17.7	12.7	21.3	16.5	11.7	19.9	59.6	25.2	55.1	23.8	51.4	22.9	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -23.8	(f) 29.4	(g) 4.5	(h) 2.7	(i) -27.0	(j) 31.3	(k) -29.6	(l) 32.8	(m) -32.2	(n) 34.3	(o) -35.4	(p) 36.3		
(4) N/A	(5) N/A	(6) N/A	(e) -24.0	(f) 21.7	(g) 4.5	(h) 1.9	(i) -27.2	(j) 23.1	(k) -29.8	(l) 24.2	(m) -32.3	(n) 25.2	(o) -35.6	(p) 26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-5.5	-3.9	-1.2	4.7	10.7	14.7	17.8	16.0	11.5	5.4	2.2	-1.7
	DBStd	8.87	6.20	5.90	4.61	3.21	4.21	3.23	3.03	2.77	3.15	3.64	4.26	5.31
	HDD10.0	2188	481	390	348	162	42	3	0	0	20	147	234	362
	HDD18.3	4574	739	623	606	410	238	117	48	83	204	401	484	620
	CDD10.0	707	0	0	0	2	63	144	241	187	66	4	0	0
	CDD18.3	51	0	0	0	0	2	8	31	11	0	0	0	0
	CDH23.3	544	0	0	0	0	46	101	314	82	1	0	0	0
	CDH26.7	113	0	0	0	0	5	16	74	17	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	657	55	42	33	33	32	56	70	82	50	69	65	60
	PrecMax	777	91	100	88	55	69	100	143	171	136	158	157	103
	PrecMin	540	7	1	3	3	6	11	5	11	9	13	3	4
	PrecStd	58	23	25	19	15	18	22	35	41	31	35	37	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	5.6	10.5	19.7	27.3	28.6	31.0	29.1	21.9	14.8	10.6	8.5
		MCWB	3.9	3.6	5.3	10.5	15.9	18.5	19.8	19.9	16.7	12.3	9.7	7.4
	2%	DB	3.8	4.2	7.9	16.5	24.4	26.2	28.7	25.3	19.7	13.0	9.6	6.7
		MCWB	3.1	3.0	4.4	9.0	14.7	17.0	19.7	18.4	15.5	11.4	9.0	5.8
	5%	DB	2.7	3.2	5.9	14.0	22.3	24.0	26.7	23.3	18.4	12.0	8.3	5.5
		MCWB	2.2	2.5	3.4	7.9	13.9	16.5	19.3	17.4	14.5	10.7	7.6	4.7
	10%	DB	1.7	2.4	4.2	11.6	19.8	21.8	24.7	21.8	17.0	11.1	7.5	4.4
		MCWB	1.2	1.8	2.3	6.4	12.8	14.9	18.6	16.5	14.3	9.9	6.8	3.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.2	6.2	11.2	17.9	20.2	22.6	21.1	17.3	13.2	10.0	7.9
		MCDB	4.8	5.0	8.9	17.7	24.2	25.4	27.5	25.8	20.1	14.0	10.4	8.2
	2%	WB	3.3	3.2	4.8	9.7	16.4	18.6	21.4	19.6	16.5	12.1	9.0	6.2
		MCDB	3.8	3.9	7.0	15.5	21.9	24.8	26.3	23.3	19.0	12.7	9.5	6.7
	5%	WB	2.3	2.6	3.7	8.3	14.8	17.2	20.2	18.4	15.5	11.1	7.8	4.9
		MCDB	2.7	3.2	5.5	13.2	20.6	22.5	24.9	22.0	17.6	11.7	8.2	5.5
	10%	WB	1.4	1.9	2.7	7.0	13.2	16.0	19.0	17.5	14.6	10.1	6.8	3.8
		MCDB	1.7	2.5	4.0	10.7	19.0	20.4	23.8	20.7	16.4	10.8	7.4	4.3
Mean Daily Temperature Range	5% DB	MDBR	5.9	6.2	8.0	10.2	11.3	9.9	10.2	9.5	8.5	7.0	4.3	4.8
		MCDBR	3.7	5.2	9.0	15.5	16.0	13.6	13.4	12.2	10.6	7.6	4.6	5.0
		MCWBR	3.6	4.4	6.0	9.0	7.8	6.6	6.0	6.3	6.7	6.0	4.4	4.7
	5% WB	MCDBR	3.8	4.6	7.7	13.5	13.7	11.8	11.3	9.7	9.0	6.1	4.7	4.8
		MCWBR	3.8	4.0	5.6	8.3	7.3	6.3	5.8	5.4	6.1	5.5	4.5	4.6
Clear-Sky Solar Irradiance	taub	0.273	0.310	0.333	0.372	0.367	0.361	0.388	0.380	0.350	0.332	0.302	N/A	
	taud	2.136	2.205	2.284	2.321	2.400	2.442	2.388	2.406	2.481	2.399	2.212	N/A	
	Ebn at Noon	473	672	789	823	859	870	832	809	775	653	435	N/A	
	Edh at Noon	42	71	90	105	104	102	105	96	75	59	40	N/A	
All-Sky Solar Radiation	RadAvg	0.24	0.81	2.31	3.88	5.30	5.55	5.23	4.12	2.53	1.10	0.31	0.15	
	RadStd	0.04	0.16	0.28	0.39	0.52	0.44	0.51	0.43	0.32	0.18	0.06	0.02	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	-135	N/A	N/A	N/A

Nomenclature: See separate page